

APOLLO SOLAR VENTURES INC.

NUTECH Hardware & Software Solutions

NUTECH School System

portal.html

Role-Based PWA · Registration · Wallets

Application Documentation

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Contents

1. Overview

portal.html is the **role-based progressive web app** (installable, works offline in demo). One file serves four roles, each seeing only their own screens. It is the management and self-service surface — and the place where new accounts are **registered** (card + face).

2. Roles & screens

Role	Screens
Admin	Dashboard, Register, Cafeteria, Credit Requests, Attendance, People, Reports, Settings
Parent	Wallet view, Order, Load Credit (request with photo proof)
Student	My Card (wallet + attendance), Order, Settings
Personnel	My Card, Order, Salary (deduction view), Settings

3. Configuration

Top of the file (also editable from the Settings page, persisted in the browser):

```
DEMO_MODE: true           // toggle in Settings
N8N_BASE: ".../webhook/cafeteria"
ADMIN_PIN: "1234"
RFID_SOURCE: "hid"        // or "arduino"
ARDUINO_BAUD: 115200
```

4. Settings page

- **Connect to live system** — the demo ↔ live switch (Save applies it).
- **n8n webhook URL + Test connection** (pings the menu action).
- **RFID reader** — USB HID or Arduino over Web Serial, with Connect + auto-reconnect.

5. Registration

Admin → **Register**. Pick the role, fill details, capture the card, and capture the face. One submit creates the **ACCOUNT** row and the matching role row under a shared ID.

Fields by role

Role	Extra fields
Student	Section, Sex, Parent ID (optional), Enrolled this year
Personnel	Role (Teacher/Staff), Monthly salary
Parent	(none beyond the common name/ID/contact/email/address)

Card capture

The **Tap** button reads the RFID UID. In live mode the reader is **auto-armed** — just tap the card and the UID fills in (HID or Arduino). In demo mode Tap generates a sample UID.

Face enrollment (required)

Registration **requires a face**. The Face enrollment panel shows a live camera; **Start camera** → **Capture face** stores the 128-number descriptor. You cannot submit until a face is captured.

1. **Start camera** → live preview appears, button becomes “Capture face”.
2. **Capture face** → descriptor stored, panel shows “✓ Face captured”.
3. Submit → the descriptor is sent as **face** and written to ACCOUNT.FACE_BIO_ID.

Requires http://localhost or HTTPS, Chrome/Edge, and internet on first use (downloads the face models once).

6. Backend actions used

Action	Feeds / does
register-person	creates ACCOUNT + role row (writes RFID + FACE_BIO_ID)
credit-request	parent submits a top-up request with proof
credit-requests	admin lists pending requests
credit-approve	approve → wallet credited (via topup)
admin-overview	dashboard totals
client-data	a person's own history (falls back to account-info)
salary	personnel monthly cafeteria spend
menu / order / account-info	cafeteria ordering + wallet

7. Running & go-live

4. **Serve over localhost/HTTPS** (VS Code Live Server, or `python -m http.server 8080`).
5. **Settings** → **Connect to live system ON** → **Test connection** → **Save**.
6. Ensure the n8n workflow is **Active** and the **register-person** branch writes FACE_BIO_ID.

8. Notes

- **Camera/serial need a secure context** — localhost or HTTPS, never file://.
- **One reader per page** — close the POS/gate tabs if they hold the Arduino.
- **Face must reach Airtable** — the register-person node must map **face** → FACE_BIO_ID, or the gate has nothing to match.